

MATHEMATICS — CLASS X

TRIGONOMETRY

Trigonometric Ratios | Standard Angles | Complementary Angles | Identities
Unit Test | Session 2025–26 | CBSE Syllabus (Ch. 8)

1. Use of calculators is NOT permitted. Standard values: $\sin 30^\circ = 1/2$, $\cos 60^\circ = 1/2$, $\tan 45^\circ = 1$, $\sqrt{2} \approx 1.414$, $\sqrt{3} \approx 1.732$.

SECTION A — Multiple Choice Questions (1 × 10 = 10 Marks)

Choose the correct answer. Each question carries 1 mark.

- Q1.** If $\sin A = 3/5$, the value of $\cos A$ is: **[[1]]**
(a) $3/4$ (b) $4/5$ (c) $5/3$ (d) $4/3$
- Q2.** The value of $(\sin^2 60^\circ + \cos^2 60^\circ)$ is: **[[1]]**
(a) 0 (b) $1/2$ (c) $\sqrt{3}/2$ (d) 1
- Q3.** The value of $\tan 45^\circ + \cot 45^\circ$ is: **[[1]]**
(a) 1 (b) $\sqrt{2}$ (c) 2 (d) 0
- Q4.** If $\cos A = 12/13$, then $\tan A =$ **[[1]]**
(a) $5/12$ (b) $12/5$ (c) $5/13$ (d) $13/5$
- Q5.** The value of $(1 - \sin^2 \theta)$ is equal to: **[[1]]**
(a) $\sin^2 \theta$ (b) $\tan^2 \theta$ (c) $\cos^2 \theta$ (d) $\cot^2 \theta$
- Q6.** $\sin(90^\circ - A)$ is equal to: **[[1]]**
(a) $\sin A$ (b) $-\sin A$ (c) $\cos A$ (d) $-\cos A$
- Q7.** $\sec^2 \theta - \tan^2 \theta =$ **[[1]]**
(a) 0 (b) 2 (c) -1 (d) 1
- Q8.** The value of $(\sin 30^\circ \times \cos 60^\circ) + (\cos 30^\circ \times \sin 60^\circ)$ is: **[[1]]**
(a) 0 (b) 1 (c) $1/2$ (d) $\sqrt{3}/2$
- Q9.** $\operatorname{cosec}^2 A - \cot^2 A$ is equal to: **[[1]]**
(a) 0 (b) -1 (c) 2 (d) 1
- Q10.** If $\tan \theta = 1/\sqrt{3}$, then $\cos \theta$ is equal to: **[[1]]**
(a) $1/2$ (b) $\sqrt{3}/2$ (c) $1/\sqrt{2}$ (d) $\sqrt{3}$

SECTION B — Very Short Answer Questions (2 × 6 = 12 Marks)

Each question carries 2 marks. Show necessary working.

- Q11.** In a right triangle ABC, right-angled at B, if $\tan A = 1/\sqrt{3}$, find the value of $\sin A \cdot \cos C + \cos A \cdot \sin C$. **[[2]]**
- Q12.** Evaluate: $(\sin^2 63^\circ + \sin^2 27^\circ) / (\cos^2 17^\circ + \cos^2 73^\circ)$. **[[2]]**
- Q13.** If $\sin \theta + \cos \theta = \sqrt{2}$, find the value of $\sin \theta \cdot \cos \theta$. **[[2]]**
- Q14.** Prove that $(1 - \cos^2 A) \cdot \operatorname{cosec}^2 A = 1$. **[[2]]**
- Q15.** Find the value of: $(\tan^2 45^\circ - \cos^2 60^\circ) / (\sin^2 30^\circ + \cos^2 30^\circ)$. **[[2]]**

Q16. If $\sin A = \cos A$, find the value of $2 \tan^2 A + \sin^2 A - 1$. **[[2]]**

SECTION C — Short Answer Questions – I ($3 \times 5 = 15$ Marks)

Each question carries 3 marks. Internal choices are indicated.

Q17. Prove the identity: $\sin \theta / (1 - \cos \theta) + \sin \theta / (1 + \cos \theta) = 2 \operatorname{cosec} \theta$. **[[3]]** OR Prove: $(\tan \theta + \sec \theta - 1) / (\tan \theta - \sec \theta + 1) = (1 + \sin \theta) / \cos \theta$

Q18. If $\sec \theta + \tan \theta = p$, prove that $\sin \theta = (p^2 - 1) / (p^2 + 1)$. **[[3]]**

Q19. Evaluate without using trigonometric tables: **[[3]]** $(\sin^2 25^\circ + \sin^2 65^\circ) / (\cos^2 24^\circ + \cos^2 66^\circ) + \sin^2 37^\circ \cdot \operatorname{cosec}^2 53^\circ$ OR Prove: $(\operatorname{cosec} A - \sin A)(\sec A - \cos A) = 1 / (\tan A + \cot A)$

Q20. Prove that: $\sqrt{[(\sec A - 1)/(\sec A + 1)]} + \sqrt{[(\sec A + 1)/(\sec A - 1)]} = 2 \operatorname{cosec} A$. **[[3]]** OR
If $\tan A = n \cdot \tan B$ and $\sin A = m \cdot \sin B$, prove that $\cos^2 A = (m^2 - 1) / (n^2 - 1)$.

Q21. If $\sin \theta - \cos \theta = 0$, find the value of $\sin^4 \theta + \cos^4 \theta$. **[[3]]** OR Evaluate: $(\sin 35^\circ / \cos 55^\circ) + (\cos 55^\circ / \sin 35^\circ) - 2 \cos 60^\circ \div \sin^2 30^\circ$.

SECTION D — Short Answer Questions – II ($4 \times 4 = 16$ Marks)

Each question carries 4 marks. Internal choices are provided.

Q22. Prove the trigonometric identity: **[[4]]** $(\sin A + \operatorname{cosec} A)^2 + (\cos A + \sec A)^2 = 7 + \tan^2 A + \cot^2 A$
OR Prove: $(\sin A - 2\sin^3 A) / (2\cos^3 A - \cos A) = \tan A$

Q23. Prove the identity: $(1 + \tan^2 A) / (1 + \cot^2 A) = \tan^2 A$. **[[4]]** OR If $\sin \theta + \sin^2 \theta = 1$, prove that $\cos^2 \theta + \cos^4 \theta = 1$.

Q24. Given that $\tan(A + B) = \sqrt{3}$ and $\tan(A - B) = 1/\sqrt{3}$, where $0^\circ < A + B \leq 90^\circ$ and $A > B$, find A and B. Hence evaluate: $\sin(A + B) \cdot \cos(A - B) + \cos(A + B) \cdot \sin(A - B)$. **[[4]]**

OR If $\cos A + \sin A = \sqrt{2} \cos A$, prove that $\cos A - \sin A = \sqrt{2} \sin A$. Hence find the value of $\tan A$.

Q25. Prove the identity: $(\sin A + \cos A)(\tan A + \cot A) = \sec A + \operatorname{cosec} A$. **[[4]]** OR Prove: $(1 + \cot A - \operatorname{cosec} A)(1 + \tan A + \sec A) = 2$.

SECTION E — Long Answer Questions ($5 \times 3 = 15$ Marks)

Each question carries 5 marks. Show all steps clearly. Internal choices are provided.

Q26. Prove the following identities: **[[5]]**

(a) $(\cos A - \sin A + 1) / (\cos A + \sin A - 1) = \operatorname{cosec} A + \cot A$, using the identity $\operatorname{cosec}^2 A = 1 + \cot^2 A$. (b) $\sin \theta / (\cot \theta + \operatorname{cosec} \theta) - \sin \theta / (\cot \theta - \operatorname{cosec} \theta) = 2$.

OR

If $\cos \theta + \sin \theta = \sqrt{2} \cos \theta$, show that $\cos \theta - \sin \theta = \sqrt{2} \sin \theta$. Also prove: $\sin^6 \theta + \cos^6 \theta = 1 - 3\sin^2 \theta \cos^2 \theta$.

Q27. Prove the following identities: **[[5]]**

(a) $\tan A / (1 - \cot A) + \cot A / (1 - \tan A) = 1 + \sec A \cdot \operatorname{cosec} A$. (b) $(\sin \theta + \cos \theta + 1)(\sin \theta - 1 + \cos \theta) \cdot \sec \theta \cdot \operatorname{cosec} \theta = 2$.